MILITARY SPECIFICATION

SULFACETAMIDE, SULFABENZAMIDE, SULFATHIAZOLE AND UREA CREAM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Sulfacetamide, Sulfabenzamide, Sulfathiazole, and Urea Cream (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals. (See 6.4)

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

FSC 6505

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be water dispersible, absorptive, vaginal cream in accordance with the following:

Sulfabenzamide	3.70%
Sulfacetamide	2.86%
Sulfathiazole	3.42%
Urea	0.64%
Cream base	qs ad 100.0%

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3.1.1 Total sulfonamides. The finished vaginal cream shall assay to contain not less than 9.00 percent and not more than 11.00 percent of total sulfonamides when assayed as specified in 4.4.1.

3.1.2 Urea. The finished Vaginal Cream shall assay to contain not less than 0.58 percent and not more than 0.70 percent of Urea, when determined, as specified in 4.4.2.

3.1.3 Accelerated aging test. All dots of finished cream are subjected to compliance with the accelerated aging test as specified in 4.4.3.

3.1.4 pH. The pH of the finished cream shall be not less than 3.00 and not more than 4.00 at 25° C. Determination shall be on the undiluted cream.

3.2 Ingredients. The ingredients used in the manufacture of the vaginal cream shall be in accordance with the following:

3.2.1 Sulfathiazole. The sulfathiazole shall be in accordance with the tests, standards, and requirements of the N.F. XI, which was official from October 1, 1960 to August 31, 1965.

3.2.2 Sulfacetamide. The sulfacetamide shall be in accordance with the tests, standards, and requirements of current N.F., including any supplements or revisions thereto.

3.2.3 Sulfabenzamide. The sulfabenzamide shall be in accordance with the following:

3.2.3.1 Assay. Sulfabenzamide shall assay not less than 99.0 percent when determined by an applicable, accurate and reproducible method.

3.2.3.2 Loss on drying. Shall be not more than 0.50 percent of its weight when dried 2 hours at 105° C.

3.2.3.3 Melting range. Shall melt within the range 180° C to 184° C when determined by the U.S.P. method class 1 materials.

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3.2.3.4 Heavy metals. Shall not be more than 20 p.p.m. when determined by an applicable, accurate and reproducible method.

3.2.4 Urea. The Urea shall be in accordance with the tests, standards, and requirements of the U.S.P., including any supplements or revisions thereto.

3.2.5 Other ingredients. All other ingredients entering into the finished vaginal cream shall be of U.S.P. or N.F. quality, or, if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade.

3.3 Contents. Filled immediate containers shall contain not less than 95 percent of the label content.

3.4 Expiration dating period. The expiration dating period shall be 36 months.

3.5 Delivery. Not less than 33 months of the expiration dating period shall remain at the time of delivery to the Government.

3.6 Workmanship. Workmanship shall be first class throughout. The material and its container shall be free from any defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period (potency period) specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

Table I. Sampling

For visual examination	Inspection level	AQL (percent defective	Unit of product
Major	S-2	1.0	Filled tube

4.2.1 For raw material and end item testing. Noncompliance with any of the following is cause for rejection.

Table II. For raw material and end item listing

Characteristic	Requirement paragraph	Inspection level	Minimum sample
Total sulfonamides	3.1.1	S-2	5
Urea	3.1.2	S-2	5
Accelerated aging test	3.1.3		
pH	3.1.4	S-2	5
Contents	3.3	S-2	10
Ingredients		**	
Sulfathiazole	3.2.1		
Sulfacetamide	3.2.2		
Sulfabenzamide			
Assay	3.2.3.1		
Loss on Drying	3.2.3.2		
Melting range	3.2.3.3		
Heavy metals	3.2.3.4		
Urea	3.2.4		

*Where possible, the same sample shall be used for two or more tests.
**Sufficient quantities of each of the raw materials shall be selected from each lot to perform tests indicated in applicable test methods. Failure of any sample to comply with test requirements shall be cause for rejection of the lot represented.

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4.3 Examination. The cream shall be examined to determine compliance with all requirements of this specification. Non-conformance will be permitted to the extent indicated in 4.2 and 4.2.1.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

	Categories and defects*
Major	
101	Cream not water dispersible.
102	Cream not absorptive, vaginal cream.
103	Cream not homogeneous, not free from separation.
104	Cream not smooth.
105	Cream not free from foreign matter.

* Inspection is not restricted to the classified possible defects listed above.

4.4 Tests.

4.4.1 Total sulfonamides.

Reagents:

1. Hydrochloric acid solution, 10% v/v.
2. 4 N Hydrochloric acid.
3. 0.1% Sodium nitrite solution (freshly prepared in purified water).
4. 0.5% Ammonium sulfamate solution (purified water).
5. 0.1% N (1-naphthyl) ethylenediamine dihydrochloride solution (purified water).

Preparation of Standard Solution:

1. Using previously dried material accurately weigh the following:

a. Sulfathiazole	171 mg.
b. Sulfacetamide	143 mg.
c. Sulfabenzamide	185 mg.

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2. Transfer to a 500 ml volumetric flask, dissolve and dilute to mark with hydrochloric acid solution.

3. Transfer a 10 ml aliquot into a 250 ml volumetric flask and dilute to mark with purified water.

Preparation of Sample Solution:

1. Accurately weigh about 2 Gram of cream into a 1 liter glass stoppered shaking bottle.

2. Add exactly 500 ml of the hydrochloric acid solution and shake for at least 15 minutes on a suitable mechanic shaking machine.

3. Filter through 41 H filter paper. Discard the first 20 ml and collect 30 - 50 ml.

4. Transfer a 10.0 ml aliquot into a 100 ml volumetric flask and dilute to mark with purified water.

Procedure:

1. Mark three 100 ml volumetric flasks as blank, standard and sample respectively.

2. To the blank flask, add 5 ml of purified water.

3. To the standard flask, add 5 ml of diluted standard solution.

4. To the sample flask, add 5 ml of sample solution.

5. To each flask, add the following:

   a. 0.5 ml 4N hydrochloric acid.

   b. 1 ml 0.1% sodium nitrite solution and allow to stand 3 minutes.

   c. 1 ml 0.5% ammonium sulfamate solution and let stand 2 minutes.

   d. 1 ml 0.1% N(1-naphthyl)ethylenediamine dihydrochloride solution.

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6. Dilute all three flasks to mark with purified water and let stand for 5 minutes.

7. Record the absorbance of the standard and the sample against the blank at 550 mu using a 1 cm absorption cell and a suitable spectrophotometer or colorimeter.

Calculations:

$$\frac{A_o \times 499 \text{ mg} \times 500 \times 10 \times 100\%}{A_s \times 500 \times 25 \times \text{Sample Wt. (mg)}} = \% \text{ Sulfonamides}$$

Where A_o = Absorbance-sample
A_s = Absorbance-standard
499 mg = Weight of sulfonamides standard

4.4.2 Urea.

Reagents:

1. Urease powder (Fisher Scientific Co., #U-21 or equivalent).
2. Cyclohexanol, reagent grade.
3. Bromcresol green indicator.
4. Methyl red indicator.
5. Boric acid, reagent grade.
6. Potassium carbonate, anhydrous, reagent grade.
7. Hydrochloric acid, reagent grade.
8. Ethanol, USP.

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Preparation of Solutions:

1. Mixed indicator solution.

   a. Mix 5 parts of 0.1% bromcresol green in ethanol with 1 part 0.1% methyl red in ethanol.

2. Boric acid solution, 4% (w/v).

3. N/10 Hydrochloric acid solution, previously standardized.

4. Urease solution, 0.3%.

   a. Transfer 300 mg of urease powder to a suitable mortar and add 4 ml of purified water.

   b. Wet and completely disperse the powder.

   c. Quantitatively transfer the dispersion to a 100 ml volumetric flask. Dilute to mark with purified water and mix thoroughly.

   d. Let stand for 30 minutes. (This solution should be used within 2 hours after preparation.)

5. Potassium carbonate solution, saturated.

Apparatus:

1. Van Slyke and Cullen type apparatus (or equivalent) for the determination of ammonia formed by the action of urease. For a duplicate determination, seven test tubes and seven aeration tube assemblies are required.

2. Water aspirator including a trap with a stopcock for controlling the air flow.

3. Constant temperature water bath, $37 \pm 2^{\circ}$ C.

4. Burrell wrist action shaker or equivalent.

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Procedure:

1. Mark seven test tubes with the numbers 1 through 7 respectively (odd numbers will be the receiver tubes).

2. Transfer about 4 grams of cream accurately weighed into the bottom of tubes 2, 4 and 6, respectively.

3. Add 25 ml of purified water to tube 6. This tube will be the reagent blank.

4. Pipet 15 ml of purified water to tubes 2 and 4.

5. Pipet 10 ml of the urease solution into tubes 2 and 4 only.

6. Seal tubes 2, 4 and 6.

7. Attach the sealed tubes to the Burrell wrist action shaker and shake the tubes at the slowest rate for 10 minutes, then at the maximum rate for an additional 20 minutes.

8. After shaking, place the tubes in the 37° C. water bath and allow to digest for 1 hour.

9. During the digestion period:

   a. Add 75 ml of the boric acid solution to each of tubes 1, 3 and 5.

   b. Add 75 ml of concentrated Sulfuric Acid to tube 7.

   c. Stopper these tubes with aeration tube assemblies.

10. After digestion is complete, cool tubes 2, 4 and 6 to room temperature.

11. Arrange the tubes in numerical order with tube 1 nearest the aspirator trap.

12. Remove the seal from tubes 2, 4 and 6 and using a minimum of purified water, rinse the seal and sides of each tube. Insert an aeration tube assembly and connect to the boric acid with latex tubing so that starting with tube 7 the outlet aeration tube is connected to the inlet of 6 etc...

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13. To each of tubes 2, 4 and 6, add 10 ml of cyclohexanol, followed by 30 ml of saturated potassium carbonate solution and immediately stopper the flask.

14. Carefully shake each digestion tube to mix the contents.

15. Open the stopcock on the aspirator trap and then turn on the water.

16. Cautiously adjust the stopcock to the fastest flow that the system will allow.

17. Aerate the system for 2 hours.

18. To stop the aeration, cautiously close the stopcock to stop the air current.

19. Remove the latex tubing from the aeration assemblies of tubes 1, 3 and 5.

20. Using a 10 ml micro-buret, titrate tubes 1 and 3 to the mixed indicator end point (disappearance of gray) with N/10 hydrochloric acid. Record the titers.

21. Dilute tube 5 to approximately the same final volume as tubes 1 and 3. Titrate to the same end point using the N/10 hydrochloric acid. This titer is the blank correction for free ammonia in the samples and reagents, and for the dilution effect on pH.

22. Calculations:

$$\frac{\text{Titer (sample-blank) x Normality x 30.03 x 100\%}}{\text{Sample weight (mg)}} = \% \text{ Urea}$$

Where: 30.03 = Equivalent weight of urea.

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4.4.3 Accelerated aging test. Select 6 tubes and subject 3 tubes to 40° C. ± 1° C. for 21 consecutive days and the other 3 tubes shall be stored at 70° - 80° F., as unaged control samples. The heated samples shall be returned to room temperature (70° - 80° F.) prior to test. All tubes shall be of the same lot number.

The cream shall then be tested in accordance with the current Defense Medical Purchase Description for assay (total sulfonamides, urea content), and pH. All results shall be within the requirements of the applicable specification except for the pH on the heated sample which may be 0.6 unit higher than the unaged pH. Thus, the pH may exceed the upper limit for the heated sample.

NOTE: The contractor, as a minimum, is required to perform the aging test on the first 2 lots and on each subsequent consecutive fourth lot for each contract. At its descretion, the Defense Personnel Support Center may test any and all lots to determine compliance with the aging test.

4.5 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored as specified in Table III. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table III. Examination of preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for Delivery shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.2.1 Unit of issue. One package containing one tube of cream and one applicator as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. Immediate container shall be a commercially available container (ointment tube) considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other. Each tube shall be supplied with a suitable measured dose plastic applicator which can be readily and securely affixed to the neck of the tube.

5.2.2.2 Unit package. The tube and applicator shall be packaged together in the suitable enclosing carton, designed with interior fittings to insure against shifting of, or damage to the contents.

5.2.2.3 Intermediate package. Six (6) units of issue shall be packaged as specified in 5.2.12.1 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. One hundred forty-four (144) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. One hundred forty-four (144) units of issue shall be packed as specified in 5.4.3.5.1 of PPP-C-186.

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5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below.

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following:

(a) labeling information in accordance with commercial practice.

(b) the national stock number.

Notes: (1) The National Stock Number shall not be preceded by any prefix.

(2) Strip labeling may be used.

(3) Label adhesion test shall not apply.

5.4.2 Intermediate package. Intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type II shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86° F.)"

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type II shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86° F.)"

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock No.	Item Identification
6505-00-299-8598	SULFACETAMIDE, SULFABENZAMIDE, SULFATHIAZOLE, AND UREA CREAM, Vaginal, 2-3/4 oz (78 Grams)

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

6.4 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superceded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1358
cs

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-S-37204 SULFACETAMIDE, SULFABENZAMIDE, SULFATHIAZOLE AND UREA CREAM

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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